Explaining and auditing models

Adult census income

LECTURE 10 LECTURE NOTES

Applications of Machine Learning

BSc Mathematics of Artificial Intelligence · Third year

Where we are

Lecture	What it built	The question it answered
3	honest evaluation; precision, recall, a chosen threshold	how good is the model, on rows it never saw?
4	logistic regression; calibration	is a probability of 0.7 right seven times in ten?
5	trees, forests, boosting	which model, and how stable is it?
7–8	networks, and the loop that trains them	how is a model with many parameters fitted?
10	**explanations and audits**	**what does it rely on, and for whom does it fail?**

OUTSIDE THE BOOK

This lecture is taught from its notes rather than from Géron, and is examined on the same terms as every other. The notes are the primary source; the deck is the lecture.

Today

1. Why look inside a model: debugging, trust, the law
2. The task: a bank's instant pre-approval, and the model behind it
3. **The method, globally** — permutation importance against the library's impurity importance; partial dependence and ICE curves
4. **The mathematics** — the Shapley value of one prediction, computed exactly
5. **The audit** — metrics per slice, three fairness criteria and why they conflict, proxies, a threshold per group
6. **Design exercise** — a digital bank's credit-limit model, audited before it is filed as a high-risk system, in nine steps

Open Lecture 10 in Google Colab

Why look inside a model that already scores well

Reason	What can go wrong unseen
Debugging	the score is high for a reason you would reject: an identifier, a column filled in after the outcome, a quirk of how the data were collected — a *shortcut* that holds in the test set and not in the world
Trust	the people who act on the model — an underwriter, a recruiter — cannot tell when to overrule it
The law	the EU AI Act classes credit scoring and recruitment as *high-risk*: data examined for bias, documentation, logging, human oversight, monitoring after release. Under the GDPR a person refused by an automated decision can ask for a human and for meaningful information about the logic
Fairness	an accuracy is an average over everyone, and the people it fails can all be in one group

✓ None of the four is answered by the test score. All of them are answered by measurements on the fitted model.

Two questions, and the tools for each

Question	About	Tool
What does the model rely on?	the whole model — *global*	permutation importance; partial dependence
	one prediction — *local*	ICE curves; Shapley values
For whom does it fail?	groups of people	metrics per slice; fairness criteria; proxies

WHAT AN EXPLANATION IS NOT

Every tool today describes *the model*: what its output does when an input changes. None of them says what would happen to a person's income if their answer changed. An explanation is a fact about a function, not about the world.

The setting: instant pre-approval at a bank

A bank's premium card needs an income above a band. Applicants answer a dozen questions in the app. A model estimates the probability that their income is above the band:

- **0.5 or above** — pre-approved on the spot
- **below** — sent to a manual check of payslips, slower; many applicants give up on it

Error	Who pays
a qualified applicant sent to the slow path	the applicant, and the bank in lost customers
an unqualified applicant pre-approved	the bank, when the documents arrive

✓ The product team removed `sex` and `race` from the model's inputs, reasoning that a model which never sees an attribute cannot use it. Today tests that reasoning.

The data, and what is kept aside

Adult census income	Value
People	48,842
Answers the model sees (categorical)	12 (6)
Fit / validation / test rows	29,304 / 9,769 / 9,769
Share of women	0.3315
Above 50K: everyone / men / women	0.2393 / 0.3038 / 0.1093

Age, work class, education twice (as a level and as years), marital status, occupation, relationship in the household, capital gains and losses, hours, country — and `fnlwgt`, the census's sampling weight.

`sex` and `race` stay beside every row, for the audit only.

A STAND-IN

The 1994 US census extract every fairness method has been tried on. The bank is a frame: the method transfers, the numbers do not.

The model the bank ships

```
model = HistGradientBoostingClassifier(categorical_features=CAT_MASK,
                                       random_state=42).fit(X_fit, y_fit)
p_test  = model.predict_proba(X_test)[:, 1]
approve = p_test >= THRESHOLD                # THRESHOLD = 0.5, the product rule
```

On the test rows	Value
Accuracy at 0.5	0.8714
Refuse everyone — the anchor	0.7607
AUC	**0.9248**
Share pre-approved	0.1988

Histogram boosting, the model a practitioner fits on a table of mixed columns; the six categorical answers declared as categories, so a code is never read as a size. A good model. The rest of the lecture is about what it does.

Which columns does the model rely on? Two ways to ask

IMPURITY IMPORTANCE — TREES ONLY

A tree grows by splits. Each split sends a node's rows left or right so that each side is **purer**: closer to all above 50K, or all below. Credit every split's gain in purity to the column it split on, weighted by the rows it split; add up over all the trees; scale so the columns sum to 1.

It asks: how much did the trees use this column to fit their rows?

PERMUTATION IMPORTANCE — ANY MODEL

Score the model on a set of rows. Then **shuffle one column**, so that every person gets someone else's value, and score it again. The fall in AUC is the column's importance: large if the predictions depend on it, zero if the model ignores it.

It asks: how much do the predictions depend on this column?

Permutation importance, defined

$$I_j \;=\; \mathrm{AUC}\big(f;\, X, y\big) \;-\; \frac{1}{R}\sum_{r=1}^{R} \mathrm{AUC}\big(f;\, X^{(\pi_r, j)}, y\big)$$

- X, y: **held-out** rows; $X^{(\pi_r, j)}$: column j replaced by a random permutation of itself
- Shuffling keeps the column's distribution and destroys its link with the target and with the rest of the row
- Needs only predictions: it works for any model, the boosted one included, at R prediction passes per column
- R repeats give a spread; report it beside the mean

```
Xp = X.copy()
Xp[:, j] = X[rng.permutation(len(X))][:, j]       # one column, or a group
drop = auc(model, X, y) - auc(model, Xp, y)
```

On which rows? The forest has memorised its own

```
forest = RandomForestClassifier(n_estimators=200, random_state=42).fit(X_fit, y_fit)
impurity = pd.Series(forest.feature_importances_, index=FEATURES)
```

TRAINING ROWS · AUC 1.0000

The rows the 200 trees were grown on. The forest gets every one right: it has memorised them. Importance here says what the forest needs to reproduce rows it has memorised.

HELD-OUT ROWS · AUC 0.9046

Rows the forest never saw. Importance here says what helps it predict people it has not seen — the bank's question.

Impurity exists only on the training rows: it is a by-product of growing the trees. Permutation runs on either. The boosted model has no `feature_importances_`, so the comparison runs on a forest.

The column to watch: `fnlwgt`

WHAT IT IS

The *final weight*: the Census Bureau's estimate of how many people in the US population this one respondent stands for.

HOW IT IS MADE

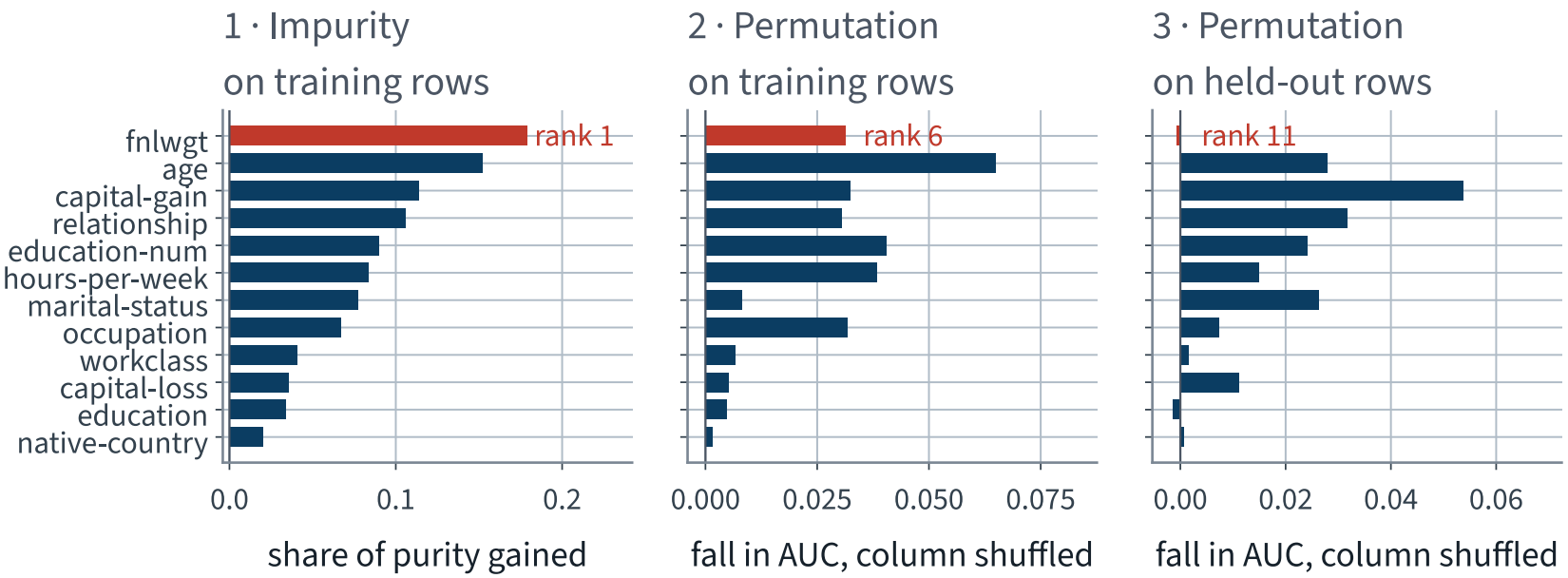
From the survey's sampling design, adjusted so that weighted counts match population totals by state, age, sex, race and Hispanic origin.

WHY IT SHOULD NOT MATTER

It records how the person was sampled, not their work, education or household. And most people have their own value: 20,077 distinct values on the fit rows.

An importance measure that ranks `fnlwgt` high is measuring something other than what predicts income.

One forest, one column, three verdicts



1 · IMPURITY, TRAINING ROWS

How much did the trees use it? `fnlwgt` **first, 0.1793.**

2 · PERMUTATION, TRAINING ROWS

What does the memorised forest need? **Still rank 6.**

3 · PERMUTATION, HELD-OUT ROWS

What predicts new people? **−0.0006: nothing.**

Only rows the model never saw show that `fnlwgt` is worth nothing.

Why impurity importance misleads: two causes, both measured

Cause	Mechanism	Measured
Computed on the training rows	a fully grown forest fits its own rows exactly, so every split, even one on noise, reduces impurity there	forest AUC on its training rows 1.0000, on test rows 0.9046
Many distinct values	a column with a value per row offers a split anywhere; deep in a tree, some split on it always cleans a small node	`fnlwgt` takes 20,077 distinct values on the fit rows; age takes 73

Panel 2 of the figure separates the two: on the training rows, permutation still credits `fnlwgt` (0.0313, rank 6), because the trees memorised it there. Only rows the model never saw say it is worth nothing.

FAILURE CONDITION — IMPORTANCE EARNED ON THE TRAINING ROWS

Impurity importance measures how much a tree *used* a column to fit its training rows, not how much the predictions on new rows depend on it. It is biased towards columns with many distinct values, and it exists only for trees.

How does the prediction change with age?

ICE — ONE APPLICANT'S CURVE

Take one applicant. Keep every answer they gave and change only their age: 20, 22, … 80. Ask the model each time. The predictions, joined, are that applicant's curve: *individual conditional expectation*.

PARTIAL DEPENDENCE — THE AVERAGE CURVE

Draw the ICE curve of every applicant. At each age, average their predictions. One curve: what the model predicts, on average, if everyone had that age.

WHAT EACH ONE TELLS YOU

The partial dependence gives the overall shape of the effect in one line. The ICE curves show whether everyone follows that shape — and sometimes nobody does.

A toy model, four applicants

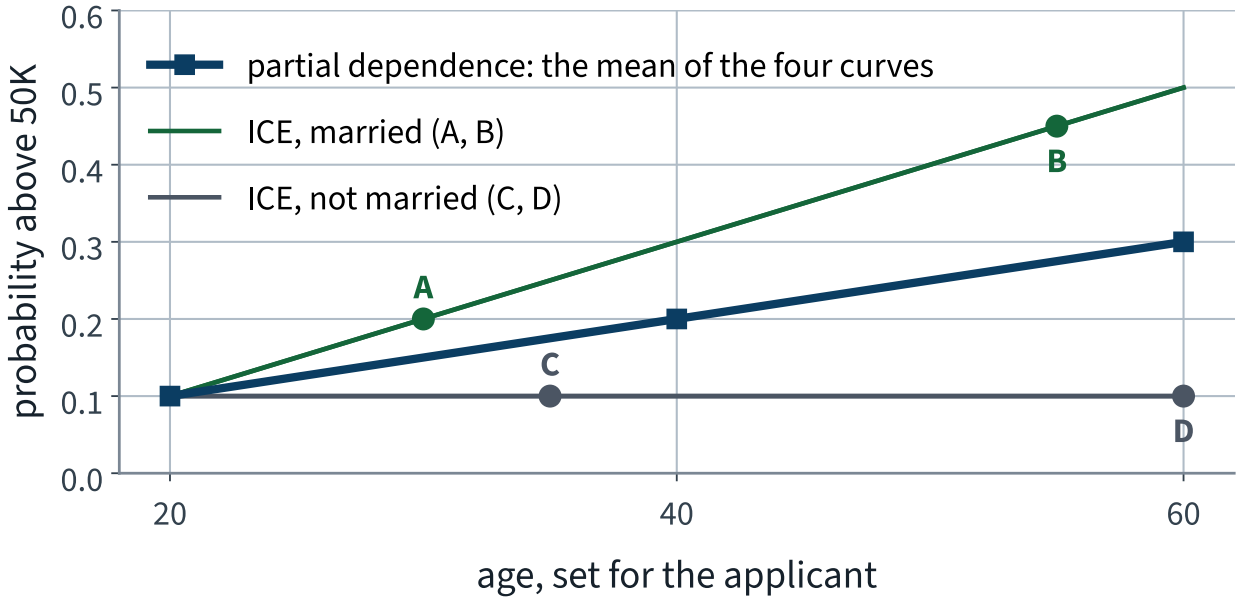
THE MODEL, WRITTEN BY HAND

Not married: 0.10 at any age. Married: 0.10 at age 20, rising by 0.10 every ten years.

Applicant	Age	Married	Own prediction	ICE at 20	at 40	at 60
A	30	yes	0.20	0.10	0.30	0.50
B	55	yes	0.45	0.10	0.30	0.50
C	35	no	0.10	0.10	0.10	0.10
D	60	no	0.10	0.10	0.10	0.10
Partial dependence (mean)				**0.10**	**0.20**	**0.30**

A's curve: set A's age to 20, 40, 60, keep "married", ask the model. A's real age only marks where A sits on the curve.

The average rises at a rate nobody has



MARRIED, A AND B

From 20 to 60: **+0.40**.

NOT MARRIED, C AND D

From 20 to 60: **0.00**.

PARTIAL DEPENDENCE

From 20 to 60: **+0.20** — true of no applicant.

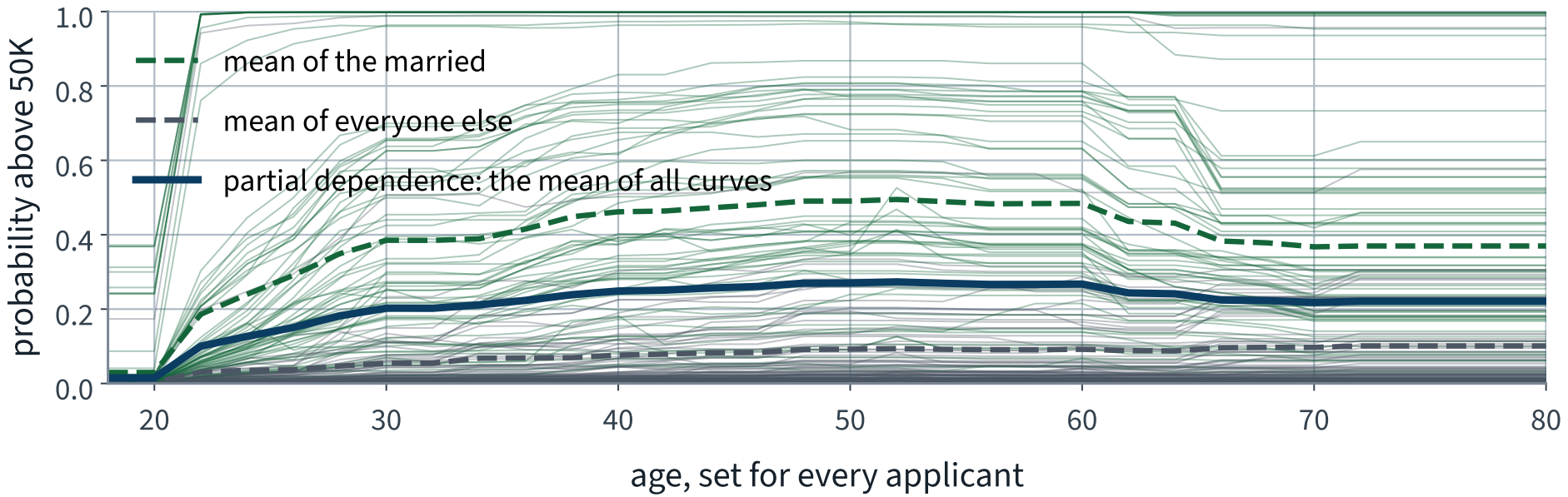
ICE and partial dependence, in general

$$\mathrm{ICE}_i(v) = f\left(v,\ \mathbf{x}_{-j}^{(i)}\right), \qquad \mathrm{PD}(v) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{ICE}_i(v)$$

- f: the model; j: the column studied (age); v: a value on the grid; $\mathbf{x}_{-j}^{(i)}$: every other answer of applicant i, kept as given
- In probabilities, the units of the product rule; on the data, 1,000 test applicants, ages 18 to 80

```
for k, v in enumerate(grid):
    Xg = X.copy(); Xg[:, j] = v                # everyone gets age v
    ice[:, k] = model.predict_proba(Xg)[:, 1]
pdp = ice.mean(axis=0)
```

The same on the data: age, for 1,000 applicants



THIN · ICE

One per applicant: green married, grey the rest.

DASHED · TWO GROUPS

The mean curve of the married and of everyone else.

THICK · PARTIAL DEPENDENCE

The mean of all the curves: like nobody.

What a partial dependence hides: an interaction

Age	Partial dependence	Married	Everyone else
20	0.015	0.030	0.003
30	0.203	0.385	0.055
50	0.270	0.490	0.092
70	0.218	0.367	0.097

The partial dependence is the weighted average of the two groups, with the share married, 0.447, as the weight. From 20 to 50 the married go from 0.030 to 0.490, everyone else from 0.003 to 0.092: the effect of age depends on another answer, and an average of the two is neither.

FAILURE CONDITION — AN AVERAGE OVER AN INTERACTION

Where the effect of one column depends on another, the partial dependence averages curves of different shapes, and can even average a rise and a fall into a flat line. Draw the ICE curves; colour them by the suspect.

One applicant asks why

The rule for choosing him: of the 826 test applicants who earn above 50K and were refused, the one at the median probability. A typical refusal, not an extreme one.

Answer	Value
age	50
marital status, relationship	married, husband
education	HS-grad, 9 years
occupation, work class	*left blank*
capital gain, loss	none
hours, country	40, United States

Probability **0.2880**, under the threshold of 0.5.

He was refused instant pre-approval, and the law entitles him to ask what in his answers counted against him.

✓ We need a number per answer that says how much it moved his probability — and that numbers add up to the probability itself.

The mathematics: the Shapley value of one prediction

Features are players; a **coalition** S is the set whose answers are the applicant's. The others are filled in from a background sample B of applicants, and the model averaged:

$$v(S) \;=\; \frac{1}{|B|}\sum_{\mathbf{b}\in B} f\left(\mathbf{x}_S,\ \mathbf{b}_{\bar{S}}\right)$$

$$\phi_i \;=\; \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!\,(M-|S|-1)!}{M!}\left(v(S\cup\{i\})-v(S)\right)$$

- $v(S\cup\{i\})-v(S)$: what revealing answer i adds to a coalition that already holds S
- The weight is the share of the $M!$ orders of revealing the answers in which exactly S comes before i: ϕ_i is the marginal contribution averaged over every order

Efficiency, and the three other properties

$$\sum_{i=1}^{M} \phi_i \;=\; v(N) - v(\varnothing) \;=\; f(\mathbf{x}) \;-\; \frac{1}{|B|} \sum_{\mathbf{b} \in B} f(\mathbf{b})$$

Property	Statement
Efficiency	**they add up to the prediction minus the baseline: in every order the contributions telescope**
Symmetry	answers that add the same to every coalition get the same value
Dummy	an answer that adds nothing to any coalition gets zero
Additivity	the values for $f + g$ are those for f plus those for g

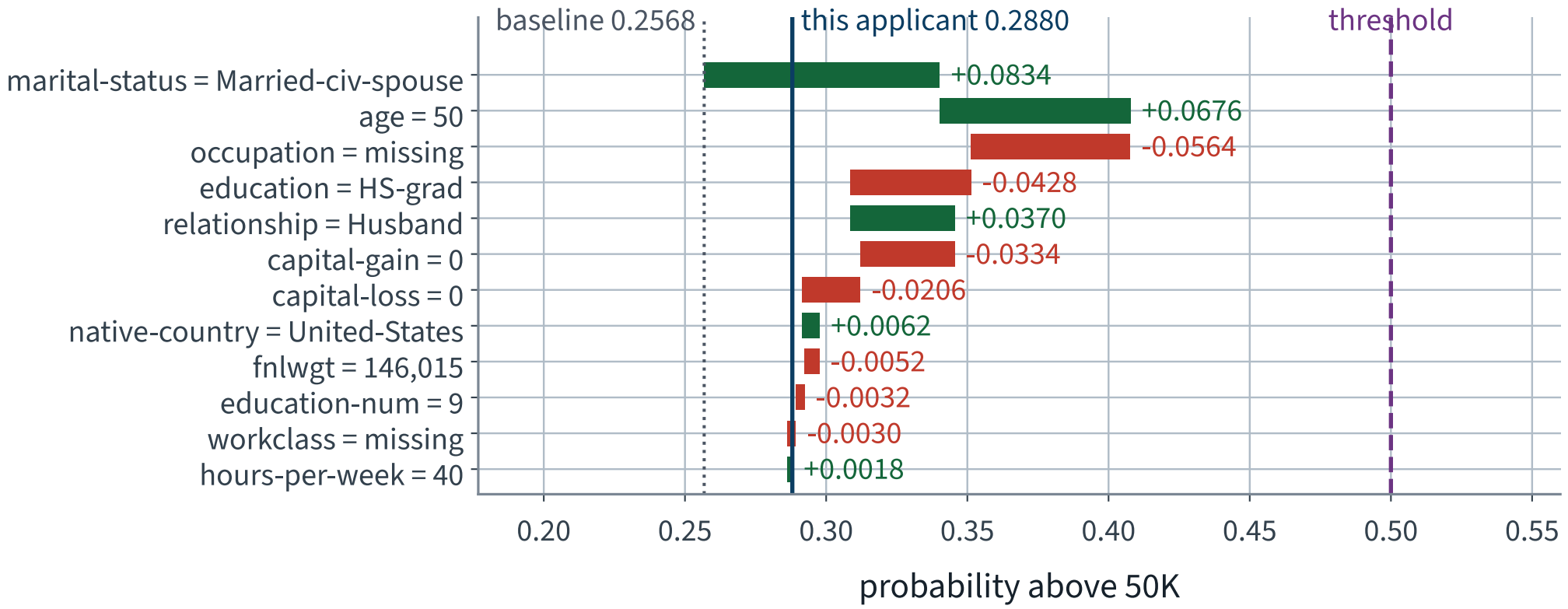
The only attribution with all four. Efficiency is proved in the notes, and examinable.

Computed exactly: every coalition, once

```
masks = np.array(list(itertools.product([False, True], repeat=M)))  # 2^M
rows  = np.where(masks[:, None, :], x, background[None])            # 2^M x |B| x M
v     = predict(rows.reshape(-1, M)).reshape(len(masks), -1).mean(1)
index, size = masks @ bit, masks.sum(1)                             # S -> row of v
for i in range(M):
    without = ~masks[:, i]                                          # every S without i
    phi[i] = (WEIGHT[size[without]]
              * (v[index[without] + bit[i]] - v[without])).sum()
```

- $M = 12$ answers: 4,096 coalitions × 200 background applicants = 819,200 predictions, in one call
- Checked first on a linear function, where the answer is known: $\phi_i = w_i \left(x_i - \bar{b}_i\right)$
- The cost doubles with every feature. At fifty, exact enumeration is out of reach: sampling estimators, and exact algorithms for trees, are what the `shap` library implements

His explanation



From the background's mean, 0.2568, to his 0.2880: the twelve values add up to 0.0312, exactly the difference.

Reading his explanation

Answer	Value	Reading
married	+0.0834	the largest push up
age 50	+0.0676	near the top of the age curve
occupation left blank	**✗ −0.0564**	**the largest thing held against him**
HS-grad	−0.0428	below the background's education
no capital gain	−0.0334	most applicants have none; the few with some score high
`fnlwgt`	−0.0052	a survey weight, moving one person's score

A blank on the form: 0.0559 of the fit rows have one, and 0.0947 of them earn above 50K against 0.2478 of the rest. The model learned that a blank predicts a lower income. Whether a blank *should* cost an applicant anything is the bank's decision — and no global summary would have put the question on the table.

Local values, averaged: the global picture again

Over 40 test applicants	Mean |Shapley|	Rank	Permutation rank
marital status	0.0823	1	2
age	0.0550	2	3
capital gain	0.0498	3	1
occupation	0.0396	4	5
relationship	0.0355	5	8
education	0.0295	6	4

The two rankings agree at the top: averaged, many local answers give back a global one.

Capital gain is third on average and moved our applicant by −0.0334: local and global answer different questions.

What each explanation cannot do, in one place

Method	Fails when	Seen here
Impurity importance	read on training rows; a column has many distinct values	`fnlwgt` first, worth nothing
Partial dependence	an interaction	married against the rest
Shapley values	the background is a choice; filled-in rows can be impossible; the cost doubles per feature	the baseline is the background's mean, 0.2568
All of them	read as causes	they describe the function, not the person

✓ Use held-out rows, draw the ICE curves, state the background — and never write "because" about the world.

The audit: the same metrics, per slice

Test rows, threshold 0.5	n	Above 50K	Base rate	Accuracy	TPR	FPR
men	6530	1984	0.3038	0.8398	0.6613	0.0823
women	3239	354	0.1093	0.9352	✗ 0.5650	0.0194
age under 25	1678	13	0.0077	0.9940	0.3846	0.0012
age 25–44	4962	1225	0.2469	0.8587	0.6163	0.0618
age 45–64	2684	1003	0.3737	0.8141	0.6899	0.1118
age 65 and over	445	97	0.2180	0.8966	0.6186	0.0259

TPR: of the applicants who earn above 50K, the share pre-approved. FPR: of those who do not, the share pre-approved.

Reading the slices: accuracy points the wrong way

- The model is **more accurate for women**, 0.9352 against 0.8398 — and a qualified woman is pre-approved **less often**, 0.5650 against 0.6613. With a base rate of 0.1093, refusing is usually right, and accuracy rewards it.
- Under 25: accuracy 0.9940, and a TPR of 0.3846 that is five people out of 13. A rate on thirteen people is an anecdote with a decimal point.

Test rows	n	Above 50K	TPR	AUC
race: White	8347	2125	0.6527	0.9214
race: Black	956	121	0.5702	0.9576
race: Asian-Pac-Islander	317	76	0.6316	0.8897
race: Other	71	6	0.5000	0.9590

✓ Read a slice by a rate that conditions on the truth — TPR, FPR — and always beside its count.

Three fairness criteria, each reasonable

$\hat{Y}$ the decision, Y the truth, A the group, s the model's probability:

Criterion	Requires, for every group a	In words
Demographic parity	$P(\hat{Y} = 1 \mid A = a)$ equal	the same share pre-approved
Equal opportunity	$P(\hat{Y} = 1 \mid Y = 1, A = a)$ equal	the same TPR: a qualified applicant's chance does not depend on the group
Calibration within groups	$P(Y = 1 \mid s, A = a) = s$	a probability means the same in every group; at a threshold, its footprint is equal PPV

PPV, from Lecture 3's precision: of those pre-approved, the share who earn above 50K.

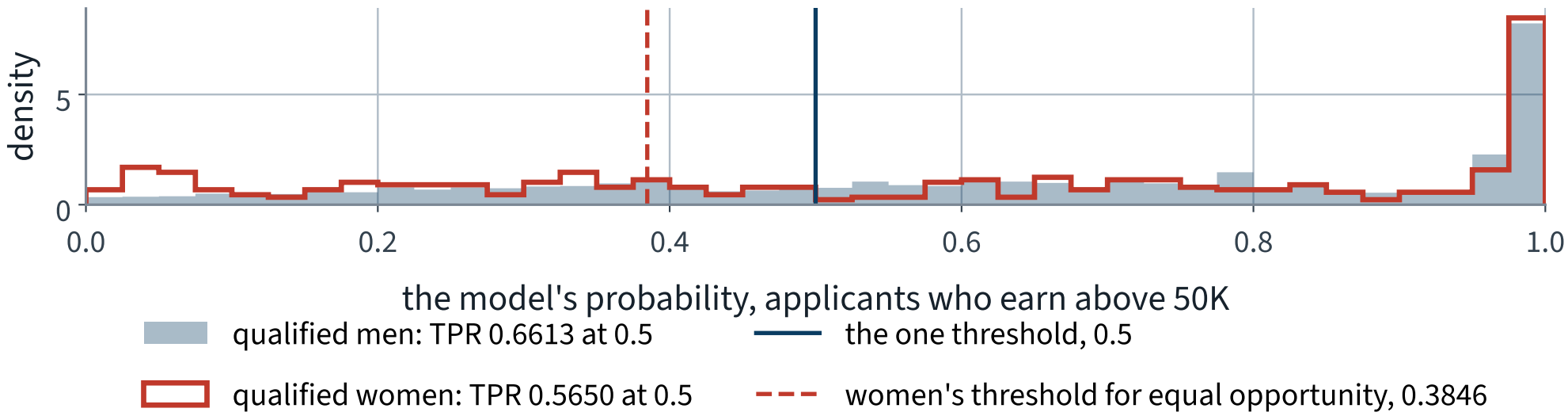
Each one has been written into a law, a policy or a court judgment somewhere. Each one protects a different person.

Measured on the shipped model

Threshold 0.5, test rows	Men	Women	Criterion
Base rate	0.3038	0.1093	
Share pre-approved	0.2582	0.0790	**✗ parity fails: ratio 0.3061**
TPR	0.6613	0.5650	**✗ equal opportunity fails: gap 0.0963**
PPV	0.7782	0.7812	**predictive parity holds ✓**
FPR	0.0823	0.0194	

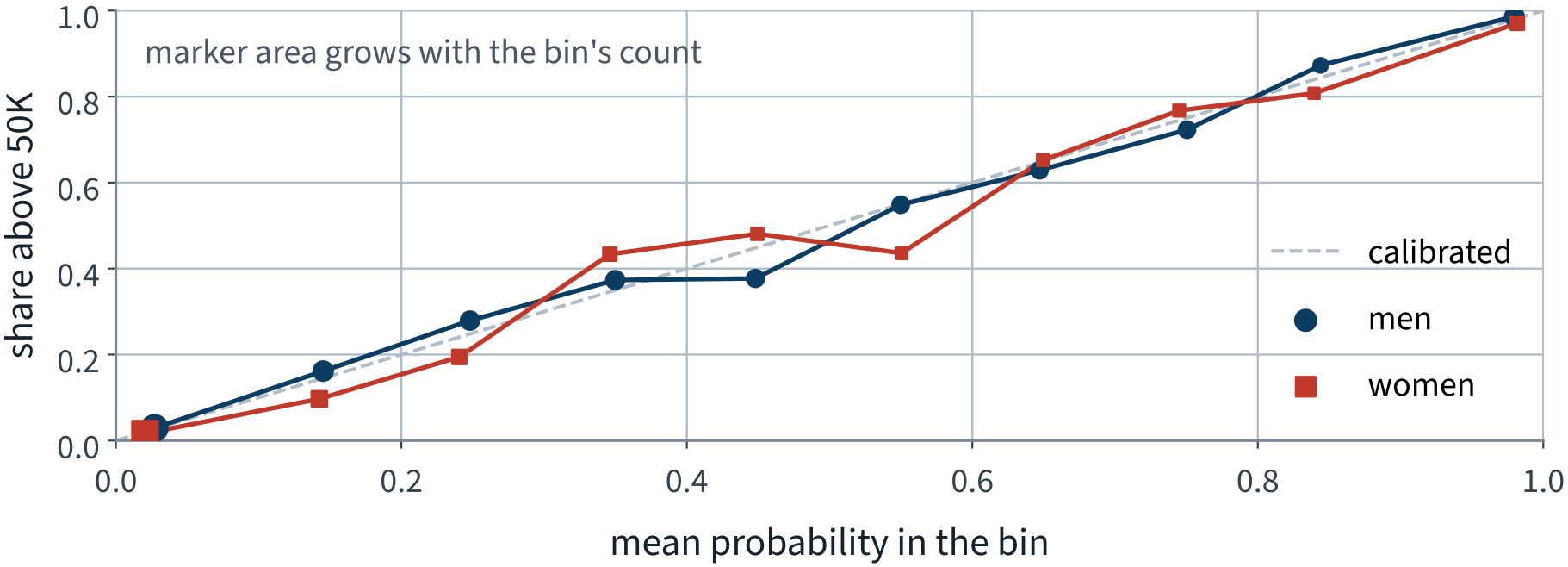
One threshold, one model, three verdicts. A pre-approval means the same thing for a man and for a woman; a qualified woman gets one less often.

Why: qualified women score lower



The model sees fewer of the answers that go with high incomes among qualified women, so more of them sit below 0.5. A threshold at 0.3846 would give them the men’s TPR on the validation rows.

Calibrated within both groups



Men: mean probability 0.3019, base rate 0.3038. Women: 0.1165 and 0.1093. Calibration error 0.0146 and 0.0137 — a probability means the same thing in both groups.

When base rates differ, they cannot all hold

For one group with base rate p, count its true positives twice — as a share of those above 50K, and of those selected:

$$\mathrm{TPR} = \frac{\text{selected} \times \mathrm{PPV}}{p}$$

- Hold any two of the three equal across groups, and the identity fixes the third by the ratio of the base rates
- Equal PPV and equal TPR force the selection ratio to $p_{\text{women}}/p_{\text{men}}$ = 0.3598. Measured, with PPV equal and TPR not: 0.3061. Parity is out of reach for any rule that keeps the other two
- Only equal base rates escape, and no model changes a base rate

WHAT THE RESULT BUYS YOU

No model satisfies every criterion. Choosing which to hold is a decision about whom to protect, and it belongs to whoever is accountable for the product.

Dropping the column does not drop the information

Fit the same model class with **sex as the target**, from the answers the bank's model sees:

Predicting sex from	Test AUC
the twelve answers	**0.9327**
without `relationship`	0.8682
without `relationship` and `marital-status`	0.8030

- `relationship` holds Husband and Wife: it names the sex of every married applicant. Hours and occupation do much of the rest.
- No single column carries it; removing columns lowers the AUC and never takes it to 0.5.

> **FAILURE CONDITION — FAIRNESS THROUGH UNAWARENESS**
>
> A model that never sees an attribute can still use it through its proxies. And a bank that never records the attribute cannot even measure whether it does: keep it, under separate access, for the audit.

Moving the threshold for one group

Men stay at 0.5; the women's threshold is chosen on the validation rows, and the test rows read once:

Test rows	One threshold	Equal opportunity	Demographic parity
Women's threshold	0.5000	0.3846	0.0783
Women pre-approved	0.0790	0.0985	0.2658
TPR, women (men 0.6613)	0.5650	**0.6525 ✓**	0.9040
PPV, women (men 0.7782)	0.7812	0.7241	**✗ 0.3717**
FPR, women (men 0.0823)	0.0194	0.0305	0.1875
Accuracy, everyone	0.8714	0.8713	0.8341

Equal opportunity: 31 more qualified women pre-approved, and 32 more who do not qualify. Demographic parity: 120 more qualified, and 485 more who do not.

A threshold per group is a policy decision

- **It uses the attribute at the moment of decision.** The model never saw sex; the rule now reads it for every applicant. Anti-discrimination law may forbid exactly that in credit — a question for the bank's legal and policy owners, not for the data team.
- **Every option has a price, and the table states it.** Equal opportunity costs almost no accuracy and lowers the women's PPV; parity costs a great deal of both.
- **There are remedies upstream.** Better answers on the form, the blank occupation fixed, a manual review for everyone in a band below the threshold — none of which needs the attribute at decision time.

> **THE DATA TEAM'S JOB**
>
> Measure every criterion, per group, with counts; show what each remedy costs and buys; write it down. The choice is made by the people accountable for it, and it is recorded with its reasons.

In practice, look out for — when explaining

Situation	What goes wrong	Do this
The library's importance chart	impurity, on training rows: an identifier or a weight comes first	permutation importance on held-out rows, with its spread
One partial dependence curve	it averages curves of different shapes	draw the ICE curves, coloured by the suspect
A Shapley value quoted alone	its background was never stated	state the background and the baseline; check that the values add up
An explanation in a customer letter	it cites a proxy, or something nobody can change	review the reasons before they are sent; map them to plain language

In practice, look out for — when auditing

Situation	What goes wrong	Do this
Accuracy per group	the group with the lower base rate looks best served	TPR and FPR per group, beside the base rate
A small slice	a rate on a dozen people swings by chance	print the count; pool, or wait for more data, before acting
“We don’t use sex”	proxies carry it; nobody can check	keep the attribute under separate access, for the audit
One fairness criterion, passed	the others fail unseen	report all three; say which one is held, and why
Thresholds tuned on the test rows	the reported equality was chosen, not measured	choose on validation, read the test rows once
An audit done once	the population drifts; a retrained model is a new model	re-audit on every retrain, and monitor slice metrics in production

A model card, in one slide

Section	For the pre-approval model
Intended use	instant pre-approval for the premium card; a refusal sends the applicant to a manual check, never to a final no
Not for	credit limits, pricing, any decision without a manual path
Data	census-style answers, 29,304 rows to fit; sex and race excluded from inputs and kept for audit
Performance	test AUC 0.9248; accuracy 0.8714 against 0.7607 for refusing everyone
By group	TPR 0.6613 men, 0.5650 women; PPV 0.7782 and 0.7812; calibrated in both
Known limits	proxies for sex (AUC 0.9327); a blank occupation lowers the score; slices under a hundred positives not assessed
Explanations	permutation importance on held-out rows; Shapley values against a stated background of 200 fit rows
Decision owner	which fairness criterion is held, and the threshold rule: named, dated, with reasons

The notebook for this lecture

Open Lecture 10 in Colab

- **Runs on free CPU**, top to bottom; it downloads the census data itself, and the numbers on these slides are its run
- The bank's model; the forest's three importances; permutation importance by hand
- ICE and partial dependence by hand, on a toy model and on the data
- Shapley values by full enumeration, checked on a linear function and by efficiency; forty applicants against the permutation ranking
- The audit: slices, three criteria and the identity, calibration per group, proxies, three threshold policies

> **WHAT TO CHANGE**
>
> In the Shapley section, draw the background from the refused applicants instead of the fit rows. The baseline drops, every value changes, and they still add up to the same probability minus the new baseline — the background is part of the explanation.

Design exercise · in the manner of an ML system-design interview

The task: audit a credit-limit model before it is filed

> **THE BRIEF**
>
> A digital bank operating across the European Union sells a credit card inside its app. The feature is the model that decides, at application, whether to offer the card and with what **credit limit**, and later whether to raise it. The users are applicants and cardholders; the business measures losses from defaults, the share of applicants who take the card, and the revenue a limit earns. Under the EU AI Act a creditworthiness model is a **high-risk system**: before it ships, the bank must document it, examine its data for bias, and put human oversight and monitoring in place.

- **Given:** application answers, the bank's own account history, credit-bureau data, and repayment outcomes for the cards it issued
- **Consumers of the output:** the app, the credit officers who review cases, the letters that tell an applicant why
- **Regulators:** the financial supervisor, the data protection authority, and anti-discrimination law

Step 1 — Frame the audit: decisions, harms, groups

- Name the decisions the model makes — offer or refuse, the limit, each raise — and the harm each error does: a good customer refused or kept at a low limit; a customer given more credit than they can repay
- Name the groups the audit will read before any number is computed: the protected attributes the law lists, age bands, regions, new customers against existing ones
- Write the metrics per harm: among customers who would repay, the share offered the card; among those who default, the share given a high limit

> **PITFALL**
>
> Auditing the overall score. A model can rank well on everyone and fail one group, and an average reported to the regulator hides exactly that. Decide the groups and the harms first, so that the numbers cannot be chosen to pass.

Step 2 — The attributes the audit needs

- The model takes no protected attribute as input. The audit needs them anyway, to measure outcomes per group
- Collect them with a legal basis and the applicant’s knowledge, store them apart from the model’s features, with access limited to the audit team and every read logged
- Measure the proxies: fit a model that predicts each attribute from the features, and list the features that carry it

PITFALL

“We do not collect it, so we cannot discriminate.” Postcode, employer, household answers and spending patterns carry the attribute, the model uses it through them, and without the attribute nobody can show whether it does. Keep it, guarded, for the audit.

Step 3 — Who was never labelled

- Repayment is observed only for applicants the old policy approved. Everyone it refused has no outcome, and they are not a random sample
- Keep a small randomised approval slice, within risk limits, so that some outcomes exist where the old policy said no; document the rest as an assumption
- Audit refusals on the applicants with outcomes, and say plainly which groups the outcomes cover thinly

> **PITFALL**
>
> Auditing only the approved. A group the old policy refused often has few labels, so its error rates rest on a handful of people — or on none — and the audit reports the model fair where it simply cannot see.

Step 4 — What the model relies on

- Permutation importance on held-out applicants, with its spread over repeats, for the technical documentation
- A column of random numbers added as a control: any feature that scores no better than it is reported as unused

> **PITFALL**
>
> Filing the library’s impurity chart. An application identifier, a timestamp or a sampling weight with thousands of distinct values comes first, the supervisor reads it as what drives the credit decision, and the bank has documented a model it does not have.

Step 5 — Shapes, and shortcuts

- ICE curves for every important feature, coloured by the suspected interaction; the partial dependence only beside them
- Look for shortcuts: a feature that encodes how the data were collected — the branch, the channel, the device — rather than whether the customer repays
- Review every important feature with the credit officers: would they defend its effect in front of a customer?

> **PITFALL**
>
> One partial dependence curve filed alone. It averages customers whose curves have different shapes, and the shape the bank documents is one the model shows nobody real.

Step 6 — Reasons for one applicant

- Shapley values for every refusal and every reduced limit, against a stated and versioned background — for instance the customers approved last quarter
- The few largest negative values become plain-language reasons in the letter, from a reviewed list; the values themselves go to the credit officer who reviews the case
- The values must add up to the score minus the baseline; that check runs on every letter

> **PITFALL**
>
> A reason that is a proxy, or that nobody can act on. “Household status” in a refusal letter tells the applicant the bank weighed something close to a protected attribute, and invites the complaint; reasons are reviewed before they can be sent, and a proxy is a finding for step seven, not a reason.

Step 7 — The audit by group

- Per group: the offer rate, the true-positive rate among customers who repay, the false-positive rate among those who default, calibration of the default probability, and the limits granted
- Every rate with its count and an interval; small groups pooled or flagged, never silently dropped
- All three fairness criteria reported, with the identity that says they cannot all hold when default rates differ

> **PITFALL**
>
> Reporting the one criterion that passes. A model calibrated in every group passes predictive parity and fails equal opportunity whenever the base rates differ, and a report that shows only the first has hidden the finding the supervisor asked for.

Step 8 — Remedies, oversight, monitoring

- Remedies upstream first: better features, a fixed form, a manual review band below the threshold where most wrongful refusals sit
- Any rule that reads a protected attribute at decision time goes to the legal and policy owners, and their decision is recorded with its reasons
- Ship with a model card, logs of every decision and its reasons, credit officers able to overrule with the override logged, and slice metrics monitored in production; re-audit on every retrain

> **PITFALL**
>
> A per-group threshold chosen by the data team. It may be unlawful differential treatment, it was never approved by anyone accountable, and when the regulator asks who decided to treat applicants differently by sex, the answer is a notebook.

Step 9 — File it: pipeline and pitfalls

Decisions, harms and groups named → protected attributes kept apart for the audit, proxies measured → the unlabelled refused accounted for → held-out permutation importance with a control → ICE and shortcut review → reviewed reasons from Shapley values → every criterion per group, with counts → remedies decided by their owners → model card, logs, oversight, monitoring.

Pitfall	Address it by
The overall score as the audit	groups and harms named before any number
“We do not collect it”	attributes kept apart, for the audit; proxies measured
Auditing only the approved	a randomised approval slice; thin groups stated
The impurity chart filed	permutation importance on held-out rows, with a control
One PDP filed alone	ICE curves, coloured by the suspected interaction
A proxy as a reason	a reviewed list of reasons, values that add up
The one criterion that passes	all three reported, with the identity
A threshold chosen by the data team	remedies upstream; the rule decided by its owners

What we did

1. Asked a good model two questions its test score cannot answer: what does it rely on, and for whom does it fail
2. Global: impurity importance ranked a survey weight first, earned on the training rows; permutation importance on held-out rows put it at zero
3. Shape: partial dependence averages ICE curves of different shapes, and can describe nobody
4. Local: the Shapley value, computed exactly over 4,096 coalitions; efficiency makes the values add up to the prediction minus the baseline; a blank on the form was the largest thing held against one applicant
5. Audit: accuracy per group points the wrong way; demographic parity, equal opportunity and calibration cannot all hold when base rates differ; dropping a column leaves its proxies; a threshold per group is a policy decision with a price the table states
6. Designed the audit of a bank’s credit-limit model before it is filed as a high-risk system, in nine steps — harms and groups first, attributes kept for the audit, the unlabelled refused, held-out permutation importance, ICE and shortcuts, reviewed reasons, every criterion per group, remedies decided by their owners — each named for the pitfall it avoids

NOT PRESENTED — FOR AFTERWARDS

Exercises Lecture 10

five, in the style of the written paper

Exercises — Lecture 10

1 A random forest ranks a census sampling weight first by impurity importance, yet shuffling that column on held-out rows does not change the AUC. Give the two reasons impurity importance can rank such a column first. [4]

2 The partial dependence of a model's probability on age rises from 0.015 at 20 to 0.270 at 50. Say what the curve averages, and how it can misrepresent the model. [4]

3 Define the Shapley value ϕ_i of feature i for one prediction, given a value function v over coalitions of the M features, and prove that $\sum_i \phi_i = v(N) - v(\varnothing)$. [5]

Solutions on Lecture 11’s deck.

Exercises — Lecture 10 (continued)

4 A classifier is applied to two groups whose base rates differ, and at its threshold the two groups have the same PPV and the same selection rate. Show that they cannot also have the same true-positive rate. [5]

5 A bank removes the applicant's sex from its model's inputs and reports that the model is therefore fair. State two things wrong with the claim, and what the bank should do instead. [4]

Solutions on Lecture 11’s deck.

THE FIVE SET AT THE END OF LECTURE 9

Solutions Lecture 9

not part of this lecture

Solutions — Lecture 9

1. A harmful-content model ends in one softmax over five harms. Say why this is the wrong output layer, and what…

✓ Harms co-occur and a softmax forces the probabilities to sum to one, so a post cannot be both violent and hateful. Use one sigmoid output per harm, each trained with its own binary cross-entropy term.

- The task is multi-label, not multi-class
- Actions are taken by a separate policy layer with a threshold per harm, so each harm needs a probability of its own

2. Harmful posts are a fraction of a percent of all posts. Explain why accuracy and the area under the ROC curve…

✓ Predicting "not harmful" for everything is almost always right, and the ROC curve looks excellent because negatives dominate the false positive rate. Report the precision–recall curve per harm, read at the precision each action requires, and check calibration.

- With rare positives a tiny false positive rate is still a flood of wrong flags; precision shows it, the ROC curve hides it
- Different harms carry different costs, so one number for all of them hides the trade-off the policy team has to make

Solutions — Lecture 9 (2)

3. The training labels are past reviewer decisions on posts the current system flagged. Name the bias, its…

✓ Selection bias: only flagged posts are ever labelled, so the next model never learns what the current one misses. Label a random sample of all traffic every week, use it for evaluation and prevalence, and add it to training.

- The loop closes on itself: the model chooses its own training data
- Only a sample drawn independently of the model can measure what the model does not see

4. Adding "number of user reports" sharply raises the offline recall of the model that checks posts…

✓ It is leakage in time: at upload the count is always zero, so the gain cannot appear in production. Compute every feature as of the moment of the decision; use reports only in a second model that re-scores the post as reactions arrive.

- A feature is valid only if the serving system can produce it at decision time
- The offline split must reproduce what is known when the decision is made

Solutions — Lecture 9 (3)

5. For each harm the policy layer has three thresholds: remove automatically, send to a human reviewer, reduce…

✓ Automatic removal: the precision the policy team accepts with no human in the loop. Review: reviewer capacity, since the expected daily volume must fit the queue. Reduced distribution: the cost of spreading a harmful post against hiding a good one. Each is a point on that harm's validation precision–recall curve, signed off as policy.

- A threshold is a decision about costs, not a hyperparameter tuned for a score
- The bands only mean something if the probabilities are calibrated

LECTURE 11 · LECTURE NOTES

Autoencoders and representation learning

A network that learns from the images alone
A linear autoencoder recovers the principal subspace, derived
The code as a representation: similar items, groups, the odd ones

Run the Lecture 10 notebook first